Tensor-Driven Procedural Generation of Cardiac Fibrosis

Theoretical Framework and Practical Pipeline Implementation

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1 Theoretical Framework (Tensor-Driven Noise)

The generation of collagen distribution relies on the Perlin noise generator $\mathcal{G}$, incorporating base fibrous obstacles, local fiber alignment, and broad spatial density variations. The general equation governing the collagen distribution field, valid for both 2D and 3D domains, is:

$$\mathcal{G}(\mathbf{p}) = \mathcal{N}_b(\mathbf{p})\Big((1-f) + f\mathcal{F}(\mathbf{p})\Big) + d\mathcal{N}_d(\mathbf{p}) \tag{1}$$

where $\mathbf{p}$ is the spatial coordinate, $\mathcal{N}_b$ is the base structural noisefield, $\mathcal{F}$ is the fibre-selecting field, and $\mathcal{N}_d$ is the density variation field.

1.1 Fractal Summation

The continuous noisefield $\mathcal{P}(\mathbf{u}, \gamma)$ utilizes a fractal sum of $N_o = 4$ octaves to generate natural roughness γ, evaluated at a mapped coordinate $\mathbf{u}$:

$$\mathcal{P}(\mathbf{u}, \gamma) = \frac{1}{2} + \frac{\sqrt{2}(1-\gamma)}{2(1-\gamma^{N_o})} \sum_{i=1}^{N_o} \gamma^{i-1} P_i(2^{i-1}\mathbf{u} + \delta\mathbf{u}_i) \tag{2}$$

1.2 Density Variation

The density variation field $\mathcal{N}_d$ is responsible for large-scale collagen heterogeneities. As these variations represent macroscopic deposition gradients rather than microscopic cellular alignment, $\mathcal{N}_d$ is evaluated globally using the patch size parameter l_d and a fixed roughness $\gamma_d = 0.5$:

$$\mathcal{N}_d(\mathbf{p}) = \mathcal{P}\left(\frac{1}{l_d}\mathbf{p},\, \gamma_d\right) \tag{3}$$

1.3 Two-Dimensional Tensor Mapping

To ensure the collagen strands organically follow the myocardial curvature, global Euclidean rotations are replaced by a spatially varying local orthogonal projector $Q(\mathbf{p})$, constructed from the local fiber vector $\mathbf{f}_0(\mathbf{p}) = (f_X, f_Y)^T$ and its transverse component $\mathbf{s}_0(\mathbf{p}) = (-f_Y, f_X)^T$:

$$Q(\mathbf{p}) = \begin{pmatrix} \mathbf{f}_0(\mathbf{p})^T \\ \mathbf{s}_0(\mathbf{p})^T \end{pmatrix} \tag{4}$$

The volume-preserving anisotropic scaling matrix $A_R = \mathrm{diag}(1/\sqrt{R}, \sqrt{R})$, controlled by the anisotropy ratio R (alignment), is applied alongside the base feature size l_b. The 2D base noisefield evaluates as:

$$\mathcal{N}_b(\mathbf{p}) = \mathcal{P}\left(\frac{1}{l_b} A_R\, Q(\mathbf{p})\, \mathbf{p},\, \gamma_b\right) \tag{5}$$

To form parallel collagenous strands along the tissue, the 2D fibre-selecting field $\mathcal{F}$ evaluates a sharpened sinusoid driven by the dot product of the spatial coordinate and the transverse biological tensor $\mathbf{s}_0$:

$$\mathcal{F}(\mathbf{p}) = \left(\frac{1}{2} + \frac{1}{2}\cos\left(\frac{2\pi}{L}(\mathbf{s}_0(\mathbf{p}) \cdot \mathbf{p}) + \Phi\mathcal{N}_f(\mathbf{p})\right)\right)^s \tag{6}$$

where L is the fibre separation distance, s is the sharpening factor, and $\mathcal{N}_f(\mathbf{p})$ provides organic phase modulation. The phase modulation field $\mathcal{N}_f$ is also transformed into the biological coordinate system to ensure the fiber "wiggliness" and dissimilarity parameters $(\beta_{\|}, \beta_{\perp})$ align with the cellular structure:

$$\mathcal{N}_f(\mathbf{p}) = \mathcal{P}\left(\begin{pmatrix}\beta_{\|} & 0 \\ 0 & \beta_{\perp}/L\end{pmatrix} Q(\mathbf{p})\,\mathbf{p},\, \gamma_f\right) \tag{7}$$

1.4 Three-Dimensional Volumetric Extension

For 3D generation, the framework utilizes the complete orthonormal basis provided by the patient-specific mesh to account for transmural rotation: the fiber $\mathbf{f}_0(\mathbf{p})$, the sheet $\mathbf{s}_0(\mathbf{p})$, and the normal $\mathbf{n}_0(\mathbf{p})$. The 3D mapping matrix $Q_{3D}(\mathbf{p})$ is:

$$Q_{3D}(\mathbf{p}) = \begin{pmatrix}\mathbf{f}_0(\mathbf{p})^T \\ \mathbf{s}_0(\mathbf{p})^T \\ \mathbf{n}_0(\mathbf{p})^T\end{pmatrix} \tag{8}$$

With the corresponding 3D anisotropic scaling matrix $A_{R(3D)}$ (incorporating the out-of-plane anisotropy factor K) and the scaled feature size $l_{b(3D)}$, the base noisefield is:

$$\mathcal{N}_b(\mathbf{p}) = \mathcal{P}\left(\frac{1}{l_{b(3D)}} A_{R(3D)}\, Q_{3D}(\mathbf{p})\,\mathbf{p},\, \gamma_b\right) \tag{9}$$

To create intricate 3D fibrosis, the fibre selection field multiplies cosines along the two transversal cellular axes ($\mathbf{s}_0$ and $\mathbf{n}_0$), creating lamellar strands of collagen that flow precisely alongside the myocytes:

$$\mathcal{F}(\mathbf{p}) = \left(\frac{1}{2} + \frac{1}{2}\prod_{j\in\{s,n\}} \cos\left(\frac{2\pi}{L_j}(\mathbf{j}_0(\mathbf{p})\cdot\mathbf{p}) + \Phi\mathcal{Z}_j(\mathbf{p})\right)\right)^s \tag{10}$$

where L_s and L_n are the lateral and normal fiber separation distances, respectively, and $\mathbf{j}_0$ maps to the corresponding transverse vectors $(\mathbf{s}_0, \mathbf{n}_0)$. The phase-modulating Perlin fields $\mathcal{Z}_j$ are evaluated on the local biological planes:

$$\mathcal{Z}_j(\mathbf{p}) = \mathcal{P}\left(\begin{pmatrix}\beta_{\|} & 0 & 0 \\ 0 & \beta_{\perp}/L_s & 0 \\ 0 & 0 & \beta_{\perp}/L_n\end{pmatrix} Q_{3D}(\mathbf{p})\,\mathbf{p},\, \gamma_f\right) \tag{11}$$

2 Gray Zone Interface Thresholding

To translate the continuous Perlin noise field into discrete binary collagen obstacles, a thresholding logic is applied. The threshold is dynamically adjusted within the Gray Zone based on a pre-computed Laplacian weight array $W \in [0, 1]$, where $W = 0$ corresponds to the True Core (maximum fibrosis) and $W = 1$ corresponds to healthy tissue (no fibrosis).

Because Perlin noise values approximate a bell-shaped distribution, applying a strict linear transition to the threshold mathematically over-penalizes the Gray Zone, forcing collagen density to drop abruptly near the core rather than decaying smoothly. To mitigate this, the framework supports two distinct decay modes that are functions of the weight W:

2.1 Linear Decay Mode

The threshold increases linearly as it approaches healthy tissue:

$$T = T_{core} + W(1 - T_{core}) \tag{12}$$

where T_{core} is the optimal threshold found via binary search to satisfy the target density within the True Core.

2.2 Exponential Decay Mode

To preserve higher collagen densities further into the Gray Zone, an exponential function is applied directly to the base threshold:

$$T = T_{safe}^{(1-W^k)} \tag{13}$$

where $k > 1$ is the control factor (greater values of k mean smoother transitions), and $T_{safe} = \max(10^{-6}, \min(0.999, T_{core}))$ acts as a mathematical shield to prevent complex number generation (e.g., negative bases). This formulation perfectly satisfies the boundary conditions ($T = T_{core}$ at $W = 0$, and $T = 1.0$ at $W = 1$) while maintaining a smoother density decay.

3 Practical Pipeline & Implementation

3.1 Data Preservation (ALG Export)

Overwriting the original tissue tags with a simple binary collagen flag destroys this vital information. Therefore, the Python bridge script (`alg_bridge.py`) preserves the original 7 (or 18) columns of the ALG mesh, leaving the tissue tag column entirely intact. The boolean collagen flag (0 or 1) is appended as a brand-new column at the extreme right of the CSV structure. The EP solver can continue reading the native tags for electrical properties while reading the final column exclusively for anatomical collagen blockades.

3.2 Implementation Control

To toggle the Gray Zone thresholding behavior for experimental runs, you can adjust the following parameters directly within the `applyFibrosisThreshold.m` script:

- `gz_decay_mode`: Set to `linear` or `power`.
- `power_k`: Value governing the penalty intensity when using the power mode (recommended $k = 2.0$).